A pool-level mass-balance inconsistency in the published Yasso20 parameter set

Working note for discussion with the Yasso development team (FMI)

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1 Summary

Using the published Yasso20 model code and the published Yasso20 calibration parameters — both taken directly from the FMI/Ryassofortran distribution — the inter-pool transfer matrix violates carbon conservation at the level of two individual donor pools. For the acid-soluble pool **A** and the non-soluble pool **N**, the fractions of decomposed carbon redistributed to other pools plus the humification fraction sum to **1.0042**, implying a respired (CO_2) fraction of **$-$0.0042** for each.

Cross-checking with Viskari et al. (2022) confirms that this arises from two deliberate calibration decisions made independently, without enforcing a joint per-column budget constraint. Every numerical result in this note is computed live from the FMI source files at document-render time.

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2 Sources

All inputs trace directly to FMI sources. No HIKET code or re-derivations are used.

Role	Source
Calibration methodology, model equations	Viskari et al. (2022), *Geosci. Model Dev.* **15**, 1735–1752, doi:10.5194/gmd-15-1735-2022
Parameter values (named)	`Yasso20_sample_parameters.rda`, FMI/Ryassofortran repository
Parameter values (raw index order)	`ParY20.dat`, same repository
Model code (R)	`Yasso20.r`, copyright 2020 FMI / Janne Pusa, GPL-3.0, same repository
Model code (Fortran)	`yassofortran.f90`, subroutine `mod5c20`, same copyright

The two parameter files are numerically identical (verified below).

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3 The published parameters

```r
# Load from the FMI .rda file
e <- new.env()
load(FMI_RDA, envir = e)
theta <- e$sample_parameters   # named numeric vector, length 35

# Cross-check against the raw ParY20.dat
raw_dat <- scan(FMI_DAT, quiet = TRUE)
stopifnot(length(raw_dat) == 35)
stopifnot(all(raw_dat == as.numeric(theta)))
message("ParY20.dat and Yasso20_sample_parameters.rda are identical: verified.")
```

Table 2: Transfer fractions and humification fraction from Yasso20_sample_parameters.rda. Highlighted rows are the source of the budget violation.

idx	rda name	Meaning	Value
5	pWA	W→A	0.5000
6	pEA	E→A	0.0000
7	pNA	N→A	1.0000
8	pAW	A→W	1.0000
9	pEW	E→W	0.9900
10	pNW	N→W	0.0000
11	pAE	A→E	0.0000
12	pWE	W→E	0.0000
13	pNE	N→E	0.0000
14	pAN	A→N	0.0000
15	pWN	W→N	0.1630
16	pEN	E→N	0.0000
31	pH	AWEN→H	0.0042

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4 The model structure

The transfer matrix is built in `Yasso20.r` exactly as follows (copied verbatim; indices refer to the FMI parameter vector `theta`):

```r
# Verbatim from FMI Yasso20.r (Janne Pusa, FMI, GPL-3.0)
row1 <- c(-1,         theta[5],  theta[6],  theta[7],  0)
row2 <- c(theta[8],  -1,         theta[9],  theta[10], 0)
row3 <- c(theta[11], theta[12], -1,         theta[13], 0)
row4 <- c(theta[14], theta[15], theta[16], -1,         0)
row5 <- c(theta[31], theta[31], theta[31], theta[31], -1)
A_p  <- matrix(c(row1, row2, row3, row4, row5), 5, 5, byrow = TRUE)
colnames(A_p) <- rownames(A_p) <- c("A","W","E","N","H")
```

The Fortran subroutine `mod5c20` in `yassofortran.f90` builds the same matrix by equivalent index-based assignments (e.g. `A(2,1) = theta(8)*ABS(A(1,1))`). Both implementations agree exactly.

5 The conservation criterion

For a donor-controlled linear compartment model $d\mathbf{x}/dt = M(\theta, c)\,\mathbf{x} + b(t)$ (Viskari et al. 2022, Eq. 1), carbon is conserved at every pool if and only if each column of the structural matrix $F(\theta)$ sums to ≤ 0, equivalently:

$$\sum_{i \neq j} p_{ij} + p_H \;\leq\; 1 \qquad \forall\, j \in \{A, W, E, N\}$$

where the sum is over all fractions leaving donor pool j to the other AWEN pools, and p_H is the humification fraction (same for all donors, row 5 of F). The remainder $1 - \sum p_{ij} - p_H$ is the fraction respired as CO_2. A negative value implies carbon creation at that pool.

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6 Result

```r
cs <- colSums(A_p)
round(cs, 6)
```

```
##       A       W       E       N       H
##  0.0042 -0.3328 -0.0058  0.0042 -1.0000
```

Table 3: Per-donor-pool carbon budget (FMI parameters). Redistributed total = transfers + p_H. A negative Respired fraction indicates carbon creation.

	Donor pool	Σ transfers out	+ p_H	Redistributed total	Respired fraction
pAW	A	1.000	0.0042	1.0042	-0.0042
pWA	W	0.663	0.0042	0.6672	0.3328
pEA	E	0.990	0.0042	0.9942	0.0058
pNA	N	1.000	0.0042	1.0042	-0.0042

Columns **A** and **N** have a redistributed total of 1.0042, implying a respired fraction of -0.0042: a pool-level violation of carbon conservation.

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7 Cause: two independent calibration decisions without a joint constraint

The origin is precisely documented in Viskari et al. (2022). Two passages, read together, explain the inconsistency.

Decision 1 — p_AW and p_NA fixed at 1 (Sect. 2.3, p. 1741):

> *"if the estimated parameter values for the p terms in Eq. (3) were within three decimals from either 0 or 1, they were set to the nearest limit value of 0 or 1, after which the calibration was redone …Parameters p_AW and p_NA were set to 1 and the other AWEN-related p values were set to 0."*

During the iterative calibration protocol, p_AW and p_NA converged to within 0.001 of 1.0 and were deliberately fixed at exactly **1** for the final calibration run.

Decision 2 — p_H calibrated independently from a separate dataset (Sect. 2.2, p. 1739):

> *"p_H (transfer fraction from AWEN pools to pool H) could have non-zero values only with the Oak Ridge global SOC dataset."*

The humification fraction p_H was estimated exclusively from the Oak Ridge global SOC measurements — a dataset entirely separate from the litter-bag data that drove the AWEN transfer fractions. The MAP value is p_H = 0.0042.

The combination: once p_AW and p_NA were fixed at 1, the subsequent independent calibration of p_H from different data had no mechanism to enforce the joint constraint p_AW + p_H $\leq$ 1. The excess is exactly p_H:

```
cat(sprintf("p_AW + p_H = %.4f + %.4f = %.4f  (excess = %.4f)\n",
            as.numeric(theta["pAW"]), as.numeric(theta["pH"]),
            as.numeric(theta["pAW"]) + as.numeric(theta["pH"]),
            as.numeric(theta["pAW"]) + as.numeric(theta["pH"]) - 1))
```

```
## p_AW + p_H = 1.0000 + 0.0042 = 1.0042  (excess = 0.0042)
```

8 The paper's stated design intent

The paper makes the design intent explicit (Sect. 2.3, p. 1741):

> *"since we assumed that only decomposition in the W pool results in CO_2, we estimated only p_EW and set p_EA to be the E pool remnant from 1 with p_EN set to 0."*

Under this assumption pools A, E, and N pass all decomposed carbon onward; only W respires. This is a biologically defensible modelling choice. The algebraically consistent implementation of this assumption requires, for any donor pool j where all outflow fractions except one are zero:

$$p_{jk} = 1 - p_H$$

so that the column sum equals exactly 1. With p_H = 0.0042, the conserving values would be:

```
pH <- as.numeric(theta["pH"])
cat(sprintf("p_AW* = 1 - p_H = 1 - %.4f = %.4f  (stored: %.4f)\n",
            pH, 1-pH, as.numeric(theta["pAW"])))
```

```
## p_AW* = 1 - p_H = 1 - 0.0042 = 0.9958  (stored: 1.0000)
```

```r
cat(sprintf("p_NA* = 1 - p_H = 1 - %.4f = %.4f  (stored: %.4f)\n",
            pH, 1-pH, as.numeric(theta["pNA"])))
```

```
## p_NA* = 1 - p_H = 1 - 0.0042 = 0.9958  (stored: 1.0000)
```

The difference between the stored value (1.0000) and the conserving value (0.9958) is exactly p_H = 0.0042 — the amount by which each column exceeds the budget.

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9 Why it was not detected

The violation is **local** (confined to pools A and N) and small (0.42 % excess). In a full simulation the carbon erroneously retained in A and N flows onward to W, which strongly respires (33.3 % of its outflow). The system as a whole remains numerically stable: a total-stock run with zero litter input and unit carbon in any AWEN pool shows declining total carbon throughout, because downstream respiration from W masks the local source. The inconsistency is only apparent from the column-level budget check shown above.

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10 Possible resolution

The joint constraint can be restored without altering the model's structural intent by replacing the fixed limit values with the algebraically consistent expressions:

$$p^*_{AW} = 1 - p_H \qquad p^*_{NA} = 1 - p_H$$

This sets the respired fraction of columns A and N to exactly zero, preserving the design assumption that "only W respires." The numerical change is 0.42 % — negligible for simulation results. The same logic applies to any future calibration where a transfer fraction converges to 1 while p_H is calibrated independently.

We would welcome the development team's view on whether this is the intended resolution, and whether the distributed parameter files should be updated.

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11 Appendix: full parameter table

Table 4: Complete Yasso20 MAP parameter vector from Yasso20_sample_parameters.rda (identical to ParY20.dat).

Index	rda name	Value
1	aA	0.5100
2	aW	5.1900
3	aE	0.1300
4	aN	0.1000

Index	rda name	Value
5	pWA	0.5000
6	pEA	0.0000
7	pNA	1.0000
8	pAW	1.0000
9	pEW	0.9900
10	pNW	0.0000
11	pAE	0.0000
12	pWE	0.0000
13	pNE	0.0000
14	pAN	0.0000
15	pWN	0.1630
16	pEN	0.0000
17	w1	0.0000
18	w2	0.0000
19	w3	0.0000
20	w4	0.0000
21	w5	0.0000
22	b1	0.1580
23	b2	-0.0020
24	bN1	0.1700
25	bN2	-0.0050
26	bH1	0.0670
27	bH2	0.0000
28	g	-1.4400
29	gN	-2.0000
30	gH	-6.9000
31	pH	0.0042
32	aH	0.0015
33	th1	-2.5500
34	th2	1.2400
35	r	0.2500

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12 Reference

Viskari, T., Pusa, J., Fer, I., Repo, A., Vira, J., and Liski, J. (2022). Calibrating the soil organic carbon model Yasso20 with multiple datasets. *Geoscientific Model Development*, **15**, 1735–1752. https://doi.org/10.5194/gmd-15-1735-2022